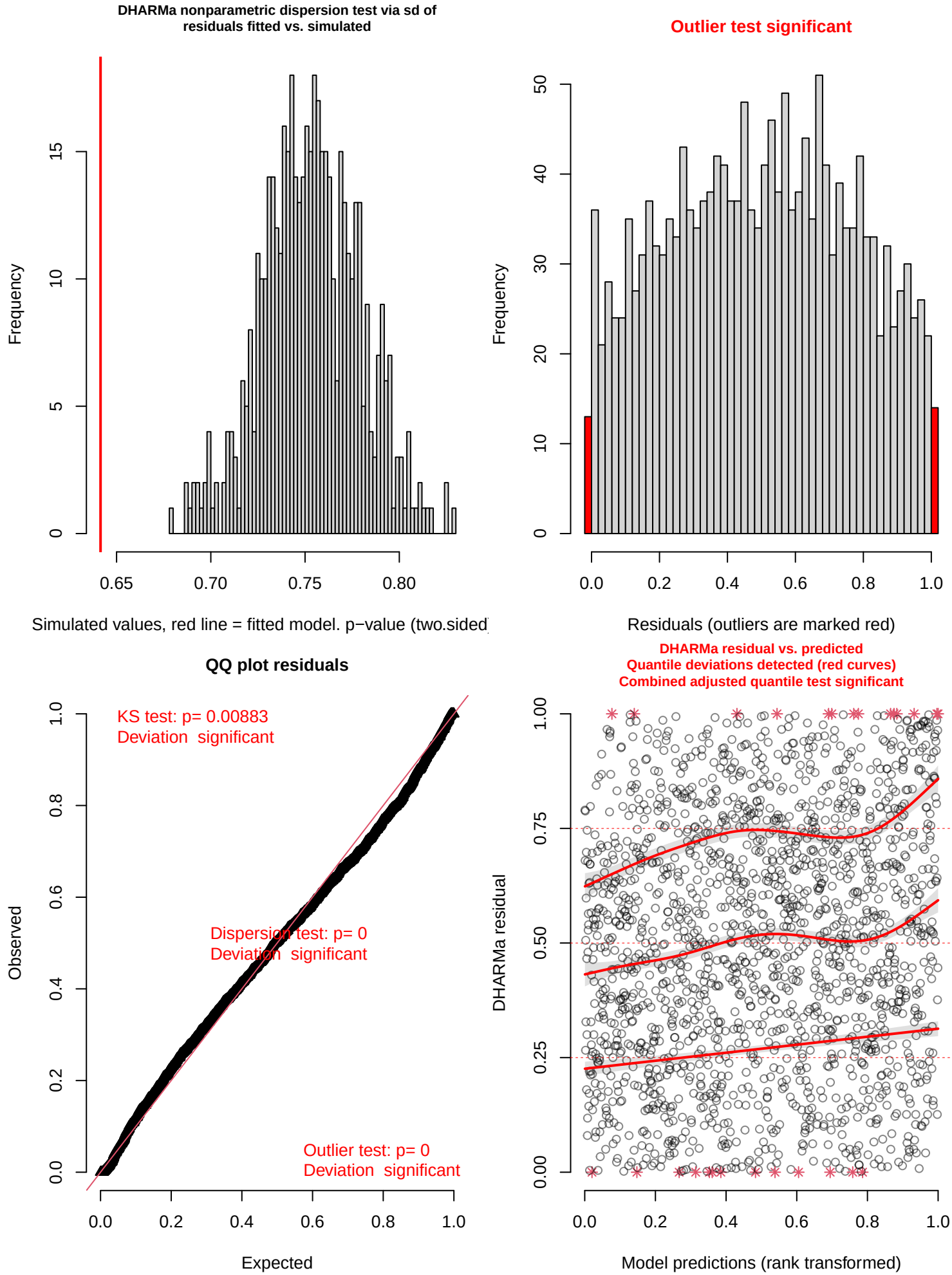
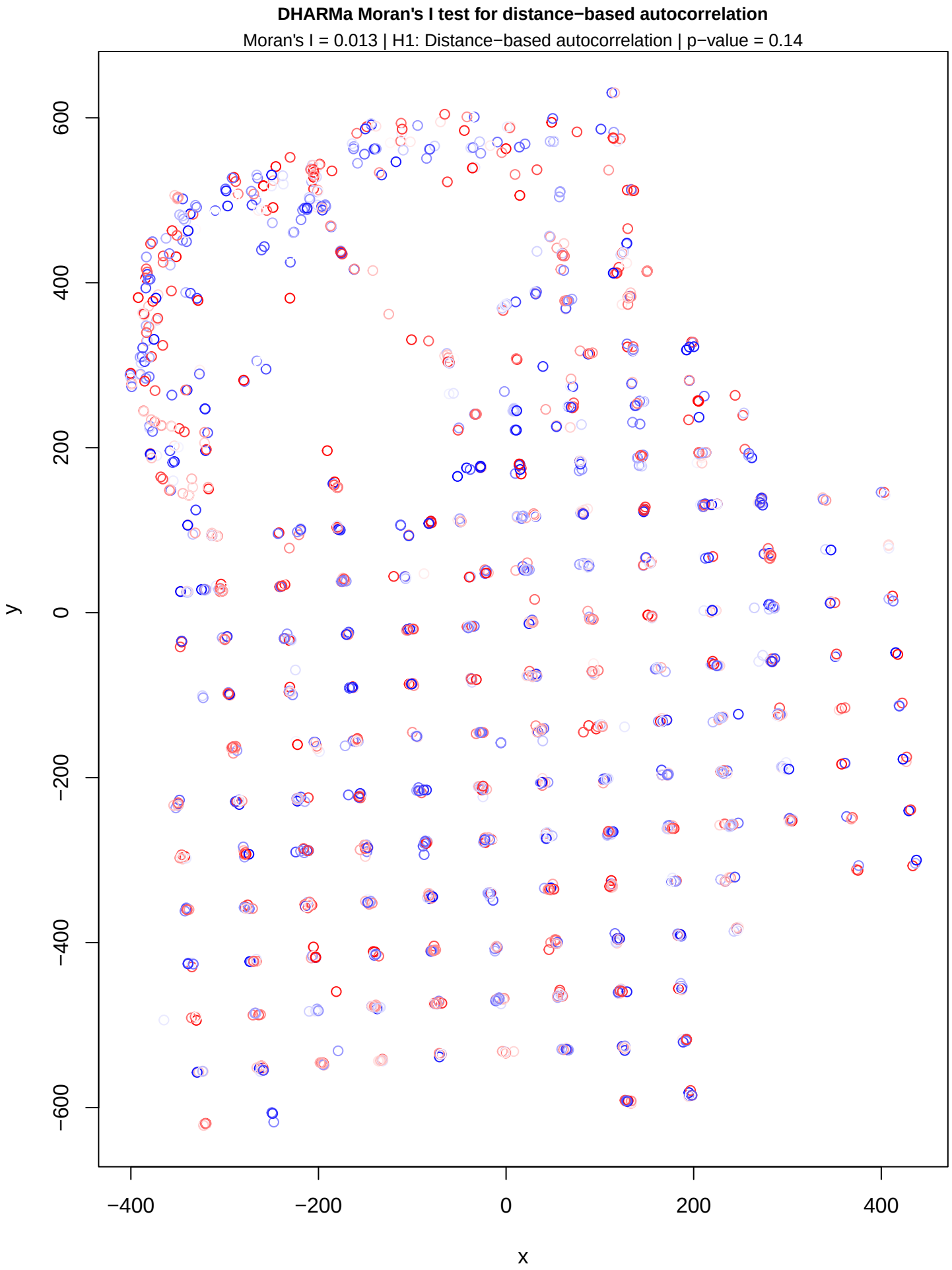
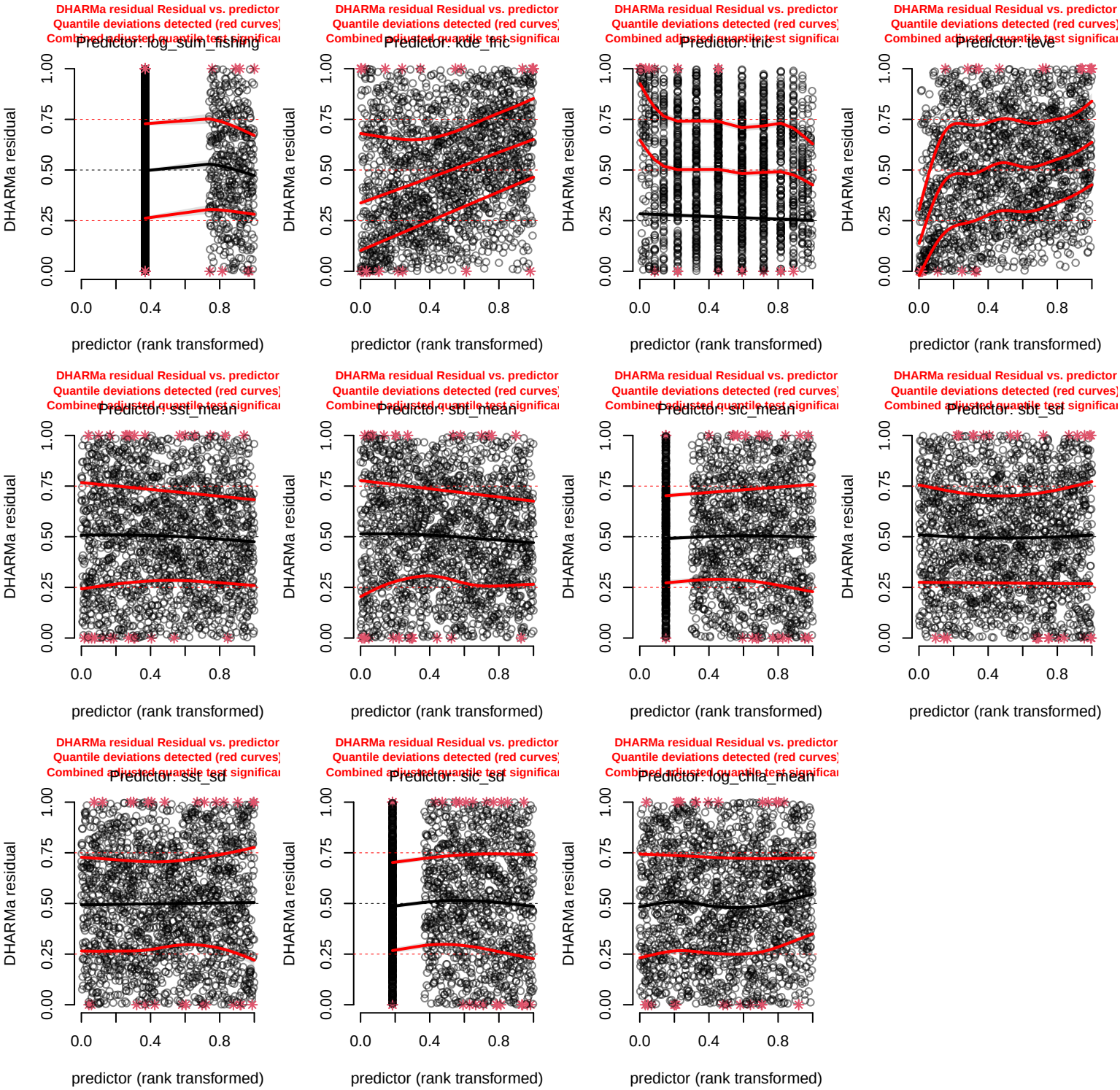
```
--------------------------------------------------------------------
MODEL SELECTION AND VALIDATION REPORT ( fishing_kde_feve model )
--------------------------------------------------------------------

## OBSERVATIONS:1755

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( fishing_kde_feve model )

Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: gaussian(link = 'identity')

Conditional model:
                coef.est coef.se
(Intercept)        −0.04    0.22
log_sum_fishing    −0.04    0.02
sbt_mean            0.02    0.07
sbt_sd             −0.08    0.04
sst_sd              0.08    0.04
sic_mean           −0.09    0.06
log_chla_mean      −0.03    0.03
depth              −0.10    0.03

Dispersion parameter: 0.84
Spatiotemporal AR1 correlation (rho): 0.88
Matérn range: 473.43
Spatiotemporal marginal AR1 SD: 0.56
REML criterion at convergence: 2313.020

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                df      AIC   delta_AIC
spatiotemporal_random_fields_ar1                12 4650.039   0.2283704
spatiotemporal_random_fields_ar1_with_anisotropy 14 4650.267  50.9142558
spatial_random_fields                           11 4701.182   0.6494775
spatial_random_fields_with_anisotropy           13 4701.831  14.6406500
spatiotemporal_random_fields_iid_with_anisotropy 13 4716.472   6.0313314
spatiotemporal_random_fields_iid                11 4722.503 152.1203297
error_only                                       9 4874.623          NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
5
10
15
0.65
0.70
0.75
0.80
Simulated values, red line = fitted model. p−value (two.sided
Outlier test significant
Frequency
0
10
20
30
40
50
0.0
0.2
0.4
0.6
0.8
1.0
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.00883
Deviation significant
Dispersion test: p= 0
Deviation significant
Outlier test: p= 0
Deviation significant
Observed
0.0
0.2
0.4
0.6
0.8
1.0
Expected
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
0.00
0.25
0.50
0.75
1.00
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.013 | H1: Distance−based autocorrelation | p−value = 0.14
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: log_sum_fishing
Predictor: kde_fric
Predictor: tric
Predictor: teve
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8